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CSAD401- OPERATING SYSTEM

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B-Tech (IT)

Analytical Problem Solving

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1. CPU Scheduling - Waiting time & Turnaround time analysis
Arrival time = 0

Process	Burst time (ms)
P1	10
P2	4
P3	6
P4	2

- (a) Calculate waiting time and turnaround time using SJF (Non-preemptive).
- (b) Compute the average waiting time and average turnaround time
- (c) Interpret whether SJF improves CPU utilisation compared to FCFS for this workload.

2. Round-Robin Scheduling - Time Quantum Impact.

Process	Burst time (ms)
P1	20
P2	5
P3	13
P4	8

Time quantum = 4ms

- (a). Construct the Gantt chart for Round Robin.
- (b). Calculate waiting time and turnaround time for each process.
- (c). Analyze how changing the quantum to 10ms would affect system performance (CPU utilization & context switching).

3. Deadlock - Safety Check using Banker's algorithm.

A system has 3 resource types (A, B, C)

Total resources : $A=10, B=5, C=7$

The Allocation & Maximum Need Matrix are :

Allocation Matrix :

Process	A	B	C
P1	0	1	0
P2	2	0	0
P3	3	0	2
P4	2	1	1
P5	0	0	2

Maximum Matrix :

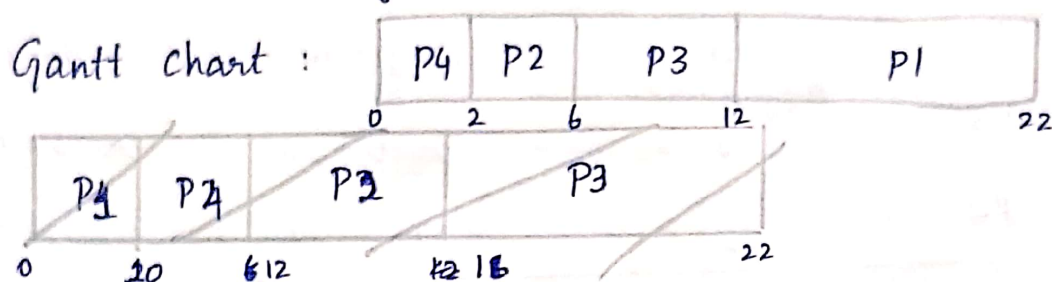
Process	A	B	C
P1	7	5	3
P2	3	2	2
P3	9	0	2
P4	4	2	2
P5	5	3	3

- Calculate the Need Matrix.
- Determine if the system is in a safe state using Banker's algorithm.
- Provide a valid safe sequence if it exists.

ANSWERS:

1. SJF Scheduling (Non-preemptive):

(a).



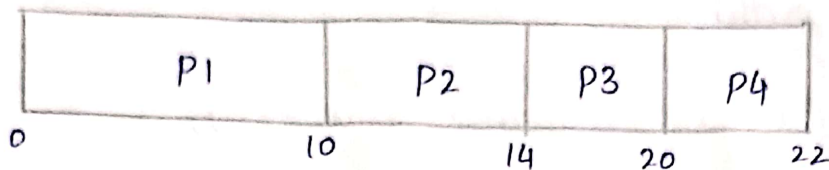
Process	Burst time (ms)	Turnaround time	Waiting time
P1	10	$22 - 0 = 22$	$22 - 10 = 12$
P2	4	$6 - 0 = 6$	$6 - 4 = 2$
P3	6	$12 - 0 = 12$	$12 - 6 = 6$
P4	2	$2 - 0 = 2$	$2 - 2 = 0$

(b). Average Turnaround time = $(22 + 6 + 12 + 2) / 4 = 42 / 4 = 10.5 \text{ sec}$

Average Waiting time = $(12 + 2 + 6 + 0) / 4 = 20 / 4 = 5 \text{ sec}$

(c). FCFS Scheduling:

Gantt chart:



Process	Burst time	Turnaround time	Waiting time
P1	10	$10 - 0 = 10$	$10 - 10 = 0$
P2	4	$14 - 0 = 14$	$14 - 4 = 10$
P3	6	$20 - 0 = 20$	$20 - 6 = 14$
P4	2	$22 - 0 = 22$	$22 - 2 = 20$

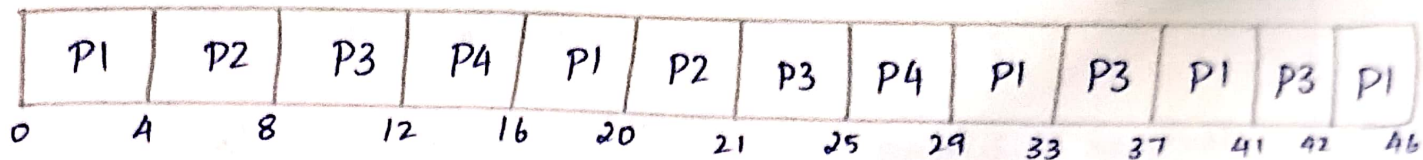
$$\text{Average Turnaround time} = (10 + 14 + 20 + 22) / 4 \\ = 66 / 4 = 16.5 \text{ sec.}$$

$$\text{Average Waiting time} = (0 + 10 + 14 + 20) / 4 \\ = 44 / 4 = 11 \text{ sec.}$$

∴ While compared to SJF and FCFS, SJF has less average turnaround time and average waiting time. So, SJF improves CPU utilisation compared to FCFS for this workload.

Gantt chart :

(a).



(b).

PROCESS	Burst time	Turnaround time	Waiting time
P1	20	$46 - 0 = 46$	$46 - 20 = 26$
P2	5	$21 - 0 = 21$	$21 - 5 = 16$
P3	13	$42 - 0 = 42$	$42 - 13 = 29$
P4	8	$29 - 0 = 29$	$29 - 8 = 21$

$$\text{Average Turnaround Time} = (46 + 21 + 42 + 29) / 4$$

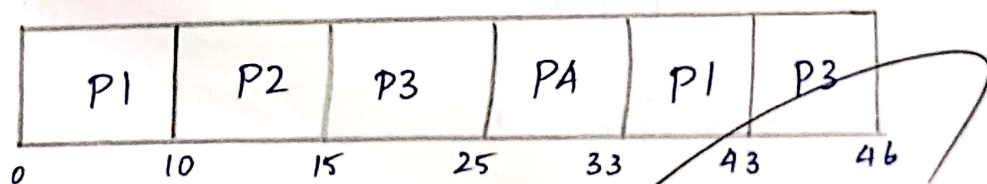
$$= 138 / 4 = 34.5$$

$$\text{Average Waiting Time} = (26 + 16 + 29 + 21) / 4$$

$$= 92 / 4 = 23$$

(c). when quantum changes to 10ms.

Gantt Chart:



Process	Burst time	Turnaround time	Waiting time
P1	20	$43 - 0 = 43$	$43 - 20 = 23$
P2	5	$15 - 0 = 15$	$15 - 5 = 10$
P3	13	$46 - 0 = 46$	$46 - 13 = 33$
P4	8	$33 - 0 = 33$	$33 - 8 = 25$

$$\text{Average turnaround time} = (43 + 15 + 16 + 33) / 4$$

$$= 107 / 4 = 26.75$$

$$\text{Average waiting time} = (23 + 10 + 33 + 25) / 4$$

$$= 91 / 4 = 22.75$$

While changing the time quantum from 4ms to 10ms it won't affect the system performance much.

3.

(a).

Need Matrix:

Process	A	B	C
P1	7	4	3
P2	1	2	2
P3	6	0	0
P4	2	1	1
P5	5	3	1

(b). Available resource = (3, 3, 2)

Safety Check:

Need $\leq$ Available

$$P_2 = (3, 3, 2) + (2, 0, 0) = (5, 3, 2) \text{ New available}$$

$$P_4 = (5, 3, 2) + (2, 1, 1) = (7, 4, 3)$$

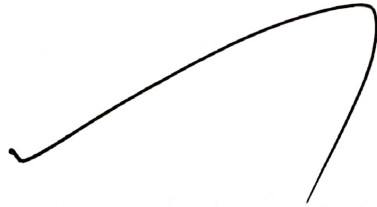
$$P_5 = (7, 4, 3) + (0, 0, 2) = (7, 4, 5)$$

$$P_1 = (7, 4, 5) + (0, 1, 0) = (7, 5, 5)$$

$$P_3 = (7, 5, 5) + (3, 0, 2) = (10, 5, 7)$$

$$P_2 \rightarrow P_4 \rightarrow P_5 \rightarrow P_1 \rightarrow P_3$$

$\therefore$ The system is in safe state.



Deadlock Detection - Resource Allocation Graph (RAG)

A system has the following resource allocations:

- * P1 is holding R1 and waiting for R2
- * P2 is holding R2 and waiting for R3
- * P3 is holding R3 and waiting for R1

- (a) Draw the Resource Allocation Graph (describe in text)
- (b) Determine whether the system is in deadlock.
- (c) Recommend a strategy (prevention (or) avoidance (or) recovery) to resolve it and justify the method.

5. CPU Utilisation - Multiprogramming Level Analysis

A system has the following process behavior:

- Each process spends 20% time in CPU and 80% waiting for I/O.
- Degree of multiprogramming = 4 processes

CPU Utilization formula:

$$U = 1 - (p^n)$$

Where:

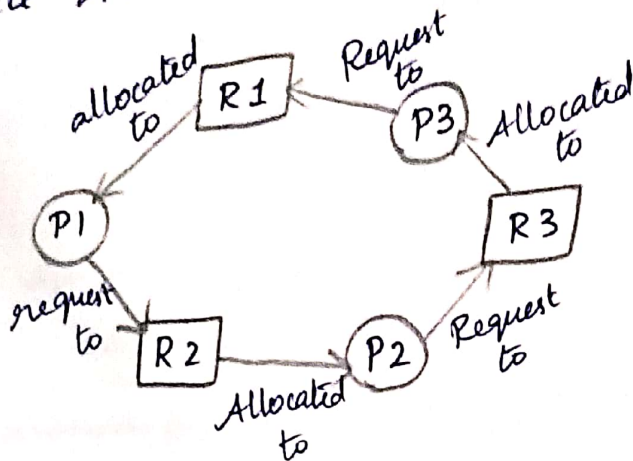
p = probability a process is waiting for I/O.

n = number of processes in memory

- (a) Calculate CPU Utilization
- (b) Compute CPU Utilization if the degree of multiprogramming increases to 6.
- (c) Interpret which configuration improves system performance and why.

ANSWERS:

Resource Allocation Graph (RAG):



$\therefore R1 \rightarrow P1 \rightarrow R2$
 $R2 \rightarrow P2 \rightarrow R3$
 $R3 \rightarrow P3 \rightarrow R1$

$\therefore$ This forms the cycle.

Deadlock Detection :

Yes, the system is in the deadlock, because this graph contains a cycle.

Recovery Method is suitable for to resolve Deadlock

Justification:

- *. Detect the deadlock using the Resource Allocation graph
- *. Abort one of the processes or preempt its resources.
- *. The released resource allows the remaining processes to continue execution.
- *. This breaks the circular wait and removes the deadlock.

CPU Utilization - Multiprogramming Level Analysis

Given:

$$\text{CPU time} = 20\%$$

$$\text{I/O waiting time} = 80\%$$

$$\text{Probability of waiting for I/O (P)} = 0.8$$

Formula:

$$U = 1 - p^n$$

(a) Calculate CPU utilization ($n=4$)

$$U = 1 - (0.8)^4$$

$$U = 1 - 0.4096$$

$$U = 0.5904$$

$$\text{CPU Utilization} = 59.04\%$$

(b) CPU Utilization ($n=6$)

$$U = 1 - (0.8)^6$$

$$U = 1 - 0.262144$$

$$U = 0.737856$$

$$\text{CPU Utilization} = 73.79\%$$

(c) Interpretation:

* For, $n=4$: CPU Utilization = 59.04%

For, $n=6$: CPU Utilization = 73.79%

→ The system with 6 processes provides better performance because the CPU remains busy more often. While some processes are waiting for I/O, other processes can use the CPU, increasing overall CPU utilization and improving system efficiency.